"SMARTCINNAMON": A MULTI-MODAL MACHINE LEARNING AND IOT FRAMEWORK FOR PRECISION CINNAMON CULTIVATION

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ABSTRACT

Ceylon cinnamon (*Cinnamomum verum*) is a high-value export crop in Sri Lanka, yet fungal leaf infections significantly reduce yield quality and plantation productivity. Existing solutions primarily detect infections at their current stage but fail to provide early warnings about disease progression, limiting farmers' ability to take timely preventive measures. This study proposes the Cinnamon Leaf Fungal Spread Predictor, a mobile-based system integrated into the SMARTCINNAMON framework. Using smartphone-captured images, the system preprocesses leaf data through automated resizing, Contrast Limited Adaptive Histogram Equalization (CLAHE), and denoising before applying a YOLOv8-seg architecture to isolate infected regions mathematically. Annotated images at multiple infection stages form a rich spatiotemporal dataset utilized to train deep recurrent networks for time-series forecasting. Long Short-Term Memory (LSTM) networks serve as the primary forecasting engine, while Gated Recurrent Unit (GRU) models are trained strictly for comparative baseline evaluation. Given a single mobile image and historical cache data, the primary LSTM model predicts both the current infection percentage and the expected infection spread over defined one-day, three-day, and seven-day horizons. Experimental results demonstrate outstanding performance, with the YOLOv8 model achieving high mAP50 scores across all disease classes, and the LSTM forecasting model delivering highly robust R^2 scores exceeding 0.94 across multiple temporal horizons. Predictions are visualized through a React Native mobile app interface, which also provides localized, actionable recommendations such as targeted fungicide spraying, canopy pruning, or plant isolation. Anticipated results include improved early detection, significantly reduced yield losses, and enhanced plantation health through proactive, data-driven disease management. In conclusion, the Cinnamon Leaf Fungal Spread Predictor introduces a novel, scalable, and farmer-friendly solution with significant potential to transform sustainable cinnamon cultivation practices.

Keywords: Cinnamon, Fungal Infection, Disease Prediction, Image Processing, Time-Series Forecasting, LSTM, GRU, Precision Agriculture, Mobile Application.

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LIST OF ABBREVIATIONS

Abbreviation		Description
LSTM	-	Long Short-Term Memory
GRU	-	Gated Recurrent Unit
CNN	-	Convolutional Neural Network
YOLO	-	You Only Look Once
ML	-	Machine Learning
DL	-	Deep Learning
AI	-	Artificial Intelligence
IoT	-	Internet of Things
RMSE	-	Root Mean Square Error
MAE	-	Mean Absolute Error
R2	-	Coefficient of Determination (R-squared)
MSE	-	Mean Squared Error
API	-	Application Programming Interface
SDLC	-	Software Development Life Cycle
IP%	-	Infection Percentage
LBP	-	Local Binary Pattern
mAP50	-	Mean Average Precision at 0.5 IoU threshold
mAP50-95	-	Mean Average Precision averaged across 0.5 to 0.95 IoU thresholds
IoU	-	Intersection over Union
WBS	-	Work Breakdown Structure
RNN	-	Recurrent Neural Network
API	-	Application Programming Interface

1. INTRODUCTION

Ceylon cinnamon (Cinnamomum verum) is one of Sri Lanka's top agricultural products that contribute significantly to the nation's economy as well as being highly regarded throughout the international spice trade [1]. Approximately 80 to 90 percent of global production of cinnamon occurs within Sri Lanka, mainly in southern and south-western parts of the country, in particular the districts of Galle, Matara, Kalutara, and Ratnapura. Ceylon cinnamon is highly sought after on an international scale, owing to its high quality, unique flavoring capabilities, low coumarin content, and numerous health benefits over cassia cinnamon types produced elsewhere.

Even though cinnamon farming generates large amounts of profit for Sri Lanka, there are several fungal diseases that threaten cinnamon cultivation. Colletotrichum gloeosporioides which causes leaf blight and anthracnose, Black Sooty Mold, which results from insect pest infestations, and Acrocercops spp. leaf miner infection all result in serious damage to cinnamon crops [1]. Leaf fungal infections have been shown to severely decrease leaf quality and reduce the export value of the leaves. Furthermore, these diseases affect the growth of bark and thus negatively impact cinnamon bark productivity and ultimately cause losses for farmers.

Conventionally, farmers growing cinnamon have used visual inspection techniques for detecting the presence of infections among leaves [1]. However, this method is both tedious and ineffective in ensuring accurate detection as infections can only be spotted at a later stage. This makes it hard to prevent infections from spreading further while causing extensive damage. Traditional disease management practices are thus not able to provide any form of prediction that could enable the implementation of measures to combat infections in good time.

Modern disease detection technology has focused on recognizing infections that have already occurred in plants rather than predicting whether a specific infection will occur within the next few days [2]. Although tools such as Plantix, Nuru, and Crop Doctor help in classifying the type of infection present using images and recommending treatments, these methods only allow farmers to identify the infections that are

currently occurring. There are no provisions for predicting infections for the future days.

In recent times, innovations in machine learning, computer vision, and mobile technologies have opened up considerable potential in improving plant disease management [4]. YOLO segmentation algorithm developed by Ultralytics allows for real-time, pixel-level disease region detection from smartphone image captures with impressive performance and minimal computing power needed. Similarly, LSTM networks, which are a type of recurrent neural network with the capacity to learn temporal dependencies of sequences of information, offer tremendous potential for predicting infection severity progression through time-series modelling [5][6][7].

The project focuses on developing the Cinnamon Leaf Fungal Spread Predictor, the independent study component of the larger SMARTCINNAMON intelligent IoT and AI framework for advanced cinnamon farming. The proposed solution uses YOLO-nano segmentation model to determine the type and degree of fungal infection in captured leaves using smartphone imagery and an LSTM time series model to predict infection spread one day, three days, and seven days in advance [3]. Additionally, GRU model is used for comparative analysis [8]. The solution is presented via a mobile app built using React Native framework, allowing for easy visualization of results and treatment advice for the farmers [9].

Figure 1:Smartphone-based image capture for cinnamon leaf fungal prediction

1.1 Background and Literature Survey

In recent years, there have been many advancements in AI and ML technologies and their applications in agriculture, specifically in managing diseases. The advancements in deep learning models, annotated image databases, and powerful mobile devices have led to the development of new AI and ML models for agricultural disease management [4]. For crops like Ceylon cinnamon, which are economically important, it is an excellent time to adopt technology that can prevent rather than cure diseases.

CNN based models have laid down the basis for plant disease detection using images. Hall and Chuan (2022) found that CNN based models trained on leaf image databases can achieve more than 95% accuracy for various crops. Some of the models used for plant disease classifications include VGG16, ResNet50, InceptionV3, and MobileNetV2, where transfer learning can make these models learn and classify the plant disease with high accuracy on even small dataset [10][11]. Although these models can classify diseases from images of leaves, it does not have the ability to predict the progress of the disease in the future.

Technological progress in object detection and instance segmentation has brought about advances in plant disease analysis that move past classification into the realm of precisely locating infected areas. Models from the You Only Look Once (YOLO) family represent the current state-of-the-art solutions in terms of real-time object detection systems [12]. In particular, YOLOv8, released in 2023, presented several innovations as compared to earlier versions, which include an anchor-free detection head, enhanced feature pyramid network architecture, and a special segmentation branch (YOLOv8-seg) capable of generating per-instance pixel masks along with bounding boxes and class labels. Thus, the introduced capability is particularly pertinent to analyzing plant diseases since it allows determining the infected area percentage.

In turn, Mithursan et al. (2024) developed an ML-based solution for identifying leaf spots and stripe cankers on cinnamon trees at Uva Wellassa University and proved the effectiveness of automatic detection of cinnamon diseases via image recognition

algorithms. The proposed solution served as proof-of-concept showing that CNN-based models could successfully classify cinnamon leaves with fungal infections with acceptable levels of accuracy in the studied setting [13]. As seen, however, it was another example of developing current-state rather than forecasting models.

The Recurrent Neural Network (RNN), along with its gated versions the Long Short-Term Memory (LSTM) network and the Gated Recurrent Unit (GRU), form the backbone architecture for the modelling of time series data [10][14]. The LSTM was invented in 1997 by Hochreiter and Schmidhuber and solves the problem of vanishing gradients experienced by standard RNNs via a combination of gates that control the retention and update of information across different time steps. This allows LSTMs to model long-term dependencies in sequences, making them suitable for modelling the progression of processes such as fungus spreading [5][6].

The GRU simplifies the architecture of the LSTM model by merging the input and forget gates into an update gate, decreasing the number of parameters and providing similar performance on most sequence prediction tasks [9]. While training faster than the LSTM, GRUs may have worse results on tasks involving capturing of long-term dependencies in sequences. For this particular study, LSTM will be used as the main predicting tool, whereas the GRU model will serve as a benchmark only [15].

Hasan et al. (2021) utilized a CNN-LSTM neural network to analyze the spatiotemporal evolution of water stress in chickpea, attaining a prediction success rate exceeding 97 percent. This study illustrated the feasibility of leveraging CNN and LSTM in a unified model for spatiotemporal analysis of plant disease progression, setting a methodology example followed in this research [5][16]. Likewise, Yao et al. (2024) designed a spatiotemporal predictive learning method based on images for the estimation of Ginkgo leaf blight development progression, which performed far better than the previous static image-based detection methods [17].

Bi et al. (2020) showed that GRU recurrent neural networks trained based on satellite time-series data could successfully identify and predict soybean sudden death syndrome with high temporal accuracy. In their study, the time series data were acquired from remote sensing imagery of soybeans at different growth stages. With

satisfactory results obtained from GRU models, this study set the basis for comparison between LSTM and GRU models used in this research for disease prediction [9].

Additionally, Kanakala and Ningappa (2025) investigated the use of LSTM and CNN architectures in multi crop disease detection, illustrating that using temporal modeling is extremely beneficial compared to static detection methods when developing disease management decisions [6]. This study also emphasized the need for carefully selecting the model's architecture and training data for achieving accurate predictions in agriculture. On the other hand, Kate (2025) went further by incorporating spatial features related to plantation with climate variables in time, using Spatio-Temporal Deep Neural Networks (ST-DNN) models to effectively predict the development of plant diseases. Kate (2025) showed that there is great potential for fusing these two types of data [7].

Integration between mobile technology and computer vision has led to the development of advanced techniques for plant disease detection at a farm level. For example, Koirala et al. (2024) analyzed the performances of several popular smartphone applications, such as Plantix, Nuru, and Crop Doctor, which offer reasonable results in terms of plant disease detection, but lack predictive abilities and are focused only on detection [3]. These limitations have driven me to design an innovative tool for predicting the spread of fungi among cinnamon leaves.

Christakakis et al. (2024) designed a mobile application based on citizen science and the application of AI to identify plant diseases and insect pests, showing how sophisticated machine learning algorithms can be implemented on mobile phones for agriculture applications [4]. Siddiqua et al. (2022) assessed the quality and weaknesses of existing mobile applications for detecting plant diseases, finding some of the common mistakes being poor identification for minorities, sensitivity to input image qualities, and lack of practical recommendations for what to do next after disease is identified [2]. Deemantha and Wickramasinghe (2025) have created an AI-based mobile application for precision cinnamon farming in Sri Lanka, noting that multilingual and offline support are essential [18].

The Roboflow platform has been established as an essential part of infrastructure used in the agricultural AI research field, allowing image data annotation, dataset versioning, augmentation pipeline design, and overall management of model training process. In the current project, an image dataset for cinnamon specific fungal disease segmentation was obtained through the Roboflow platform.

Le et al. (2024) presented research into how hybrid CNN-RNN techniques along with Liquid Time Constant Networks could be used to enhance the accuracy of plant disease diagnosis, showing that the integration of both spatial and temporal models is continuing to evolve in agriculture [12].

The literature indicates that although advances have been made in detecting plant disease, the scientific community has yet to design a mobile application for predicting the spread of fungus infections among cinnamon plants through temporal modelling.

1.2 Research Gap

However, a thorough critique of current agricultural technology trends shows that there are several major flaws with respect to the development of computational tools for managing cinnamon leaf infections. One issue is that there is an enormous gap between the problem of recognition of disease symptoms and prediction of its further development. Current approaches based on plant disease diagnosis using artificial intelligence techniques use only one-time point imaging data [10]. Such solutions are quite capable of recognizing what type of infection a leaf currently suffers from, e.g., identifying a leaf affected by necrosis as a result of infection by the Colletotrichum gloeosporioides pathogen. At the same time, such approaches do not possess necessary predictive power to warn farmers about the dynamics of infection propagation within next few days/weeks. This makes farmers operate only in a reactive manner instead of taking preventive actions [2][9][19].

Another problem is the complete lack of spatial-temporal modeling of cinnamon-specific infections within the realm of precision agriculture. Despite the presence of highly tuned artificial intelligence solutions targeting such staple crops as rice or corn on the market, there is no customized solution available for cinnamon trees [1].

It is necessary to note the exact growth rate of this fungus, lesions, and sensitivity of Stenella spp., and Colletotrichum gloeosporioides to particular environmental factors in the local microclimate of Sri Lanka; therefore, specific mathematical models need to be designed specifically for these species, and generic models of plant diseases will not suffice. One of the major issues in disease detection using ML algorithms in the case of cinnamon leaf disease is the absence of appropriate longitudinal datasets for training the algorithm [1][20].

Another problem that needs to be addressed is the lack of practical solutions for smallholder farmers. While many academic research models are based on complex hybrid CNN-RNN frameworks, most are not suitable to be implemented in practice since they are developed only theoretically and on the high performance computing servers [5][17]. However, in order to provide farmers with useful agronomic advice, it

is critical to develop models which could predict disease spread and suggest actions that need to be taken [4][18][21].

Table 1: Comparison with existing systems and identifying gaps

Reference	**Functionalities of the Application**	**Technology and techniques Used for implementation**	**Devices in which can be used**	**Accuracy**
Image-Based Cinnamon Disease Identification [13]	Static identification and stage detection of leaf spot and stripe canker on cinnamon plants.	Convolutional Neural Networks (CNN) and YOLOv5 for object detection..	PC / General Computing	Moderate
CNN-LSTM Plant Stress Detection [5]	Detection and monitoring of spatiotemporal water stress progression in plants.	Hybrid CNN-LSTM architecture combining spatial extraction and temporal sequence modeling.	High-performance computing clusters	High
Disease Progression Prediction (DPP) [17]	Mathematical forecasting of leaf blight progression based on continuous image sequences.	Spatiotemporal predictive learning, multi-level feature translation, and enhanced spatiotemporal attention.	PC / General Computing	High
Generalized Plant Apps [3]	Identification of tree species and static diseases from	Automatic Visual Recognition, using	Android and iOS mobile devices	Medium

	photographs of their leaves.	standard computer vision components.		
AIoT Smart Farm Management [22]	Plant disease detection, environment management, and LLM-powered interactive chatbot for care suggestions	Lightweight MobileViT model, transfer learning, and Internet of Things (IoT) sensors.	Android and iOS mobile devices	High
Cinnamon Leaf Fungal Spread Predictor (Our System)	Classification of plants, instance segmentation of fungal lesions, and mathematical forecasting of spatiotemporal disease spread with actionable treatments.	Computer vision (YOLO) and transfer learning based on deep CNNs, combined with Recurrent Neural Networks (LSTM primary, GRU comparative) for temporal analysis.	Android and iOS mobile devices	Higher accuracy is expected

1.3 Research Problem

The main research question that serves as a framework for the entire comprehensive report focuses on the economic and biological susceptibility of the Ceylon cinnamon industry to the fast and unanticipated development of fungal infections of its leaves [5][17][23]. Since Cinnamomum verum is planted mainly in tropical climates where the surrounding humidity levels are quite high as well as dense canopies are formed due to tree growth, fungal agents are able to reproduce exponentially in such an environment [5].

Given that farmers do not have any tools for detecting fungal symptoms except for human eyes, they automatically start working at a significant time disadvantage [2]. In other words, the absence of data about how the plant disease will continue to develop in the coming weeks makes it almost impossible for farmers to undertake preventative measures, which results in several negative consequences, including a decrease in crop yields since the amount of harvested plant matter becomes smaller, a lower level of valuable oils produced by the plant, increased expenses on applying chemical means of protection against fungi, and even rejection of shipped products in the international market due to low product quality [5][17].

1.3.1 Absence of Forecasting for Infection Progression

Mobile applications for diseases and most academic approaches focus on classification at a particular time point (disease-free vs. diseased or disease classification) rather than disease forecasting. While farmers can identify a diseased leaf easily, there is no mathematical way to predict if an infection, which covers say 18% at a given time point, will progress up to 35-40% in three days. Lack of forecasting does not enable a farmer to plan a response, such as deciding on when and how to spray chemicals or biological solutions, canopy sanitization, and even harvesting decisions. At the heart of this issue lies a well-developed time series prediction framework based on field images.

1.3.2 Lack of Cinnamon-Specific Longitudinal Data and Ground Truth

The progression models inherently necessitate time-based data involving the exact same leaf/plant observed through time, where mathematically accurate ground truthing of infection area is provided at each time point. In the case of the Ceylon cinnamon species, such datasets do not exist; any existing data involves cross-sectional observations that provide only a broad classification of either diseased or not diseased. The first step required to create such a dataset is an incredibly large one, namely to create a protocol involving sequential image acquisition in unstable mobile environments followed by effective segmentation procedures.

1.3.3 Limited Accessibility and Actionable Guidance

Many approaches for predicting agricultural diseases usually rely extensively on dense Internet of Things sensors and constant cloud-based computing resources. The reliance poses very large costs and connectivity challenges for the farmers who cultivate small plots of cinnamon in rural parts of Sri Lanka. In cases where even an accurate prediction has been made by academic algorithms, these predictions are mostly left in raw mathematical form without being transformed into useful agronomic processes. Consequently, another crucial dimension of the problem that needs to be addressed is the issue of UX; the task of delivering light inference using a mobile phone, and coupling complex spread predictions with practical local instructions.

1.4 Research Objectives

To systematically dismantle the barriers identified in the research gap and directly resolve the articulated research problem, this study outlines a comprehensive set of objectives designed to transition cinnamon cultivation from a vulnerable, reactive posture to a highly proactive, technologically empowered state.

1.4.1 Main Objective

The primary objective of this research is to engineer, validate, and deploy a mobile-integrated predictive framework designated as the Cinnamon Leaf Fungal Spread Predictor that utilizes advanced multi-modal machine learning to not only accurately detect existing fungal infections but to mathematically forecast their spatiotemporal expansion over a defined future horizon. This framework aims to empower smallholder cinnamon farmers with preemptive decision-making capabilities, thereby minimizing crop loss, reducing agrochemical dependency, and ensuring long-term agricultural sustainability.

1.4.2 Specific objectives

1. **Temporal Data Curation and Advanced Image Processing:** In order to design a robust computational pipeline for acquiring and pre-processing the acquired imagery data of the infected cinnamon leaves using advanced computer vision algorithms such as image resizing, chroma denoising through Non-Local Means algorithms, and contrast improvement using CLAHE in order to detect both necrotic and mycelia regions from the unaffected plant tissues effectively.
2. **Feature Extraction and Severity Estimation:** t aims at creating an algorithm to extract features including geometrical and textural aspects from the YOLO segmentation masks generated above. An important part of this task is the creation of a formula for calculating the Infection Percentage (IP%). The idea

is to convert pixel geometries into time-series sequences describing the temporal growth of the fungus on the leaves.

3. **LSTM and GRU Deep Learning Models:** Design, train, and optimize LSTM and GRU deep learning models. In particular, our main objective here would be developing a highly optimized Long Short Term Memory (LSTM) model to learn from historical infection rate data in order to generate predictions in terms of percentage-based infection rates in the future. In addition, a separate gated recurrent unit (GRU) model will be required to act as a comparison mathematical baseline.

4. **Multi-Horizon Evaluation:** To validate the efficacy of the temporal prediction module in consistently generating high R Squared values when forecasting infection spread over different multiple day operational horizons, namely, one day, three days, and seven days.

5. **Mobile Interface and Decision Support Implementation:** To design a user-friendly and resource-efficient React Native mobile application that graphically presents the forecast results and automatically converts the sophisticated outputs of the LSTM model into practical, location-specific farming decisions (such as exact chemical applications and pruning guidelines) that can be implemented by farmers with different digital literacy levels.

2. METHODOLOGY

The methodological framework of the Cinnamon Leaf Fungal Spread Predictor is architected as a highly sophisticated, multi-stage computational pipeline. It is specifically designed to transition seamlessly from raw, unstructured visual data captured in the chaotic field environment into highly structured, deterministic spatiotemporal forecasts. This chapter meticulously details the mathematical theories, architectural configurations, and algorithmic strategies deployed at each phase of the system, commercialization viability, and code-level implementations.

2.1 Proposed Methodology

The proposed methodology presents a structured pipeline for detecting and predicting fungal diseases in cinnamon leaves using image processing and deep learning techniques. Initially, leaf images are collected and enhanced through preprocessing and data augmentation to improve model performance. A YOLOv8-based segmentation model is then used to identify and isolate infected regions, followed by the calculation of infection severity using a pixel-based Infection Percentage (IP%) metric. To forecast disease progression over time, a Long Short-Term Memory (LSTM) network is employed using historical data patterns. Finally, the complete system is integrated into a mobile platform to enable real-time disease detection and prediction in field conditions.

2.1.1 Data Collection

In this research, cinnamon leaf images were collected to develop a fungal disease detection and prediction system. The images were captured from real-world environments using mobile devices, ensuring that the dataset reflects practical field conditions such as variations in lighting, background, and leaf orientation.

The dataset includes both healthy and infected cinnamon leaf samples. The primary classes considered in this study are:

1. Healthy Leaf
2. Colletotrichum gloeosporioides
3. Black Sooty Mold
4. Leaf Miner (Acrocercops spp.)

These classes were selected based on their prevalence and impact on cinnamon cultivation. Including healthy leaf samples allows the model to accurately distinguish between normal and infected conditions, improving classification reliability and reducing false positives.

The collected images were subsequently annotated and used for training the YOLOv8 segmentation model.

The following figures illustrate sample images from the dataset:

Figure 2: Healthy cinnamon leaf sample

Figure 3:Cinnamon leaf infected with Colletotrichum gloeosporioides

Figure 4:Cinnamon leaf affected by Leaf Miner (Acrocercops spp.)

Figure 5:Cinnamon leaf showing Black Sooty Mold infection

2.1.2 Data Augmentation and Dataset Structure

The dataset used for this study was obtained through an online annotation platform and prepared in a format compatible with the YOLOv8 segmentation framework. The dataset consists of cinnamon leaf images representing different disease conditions,

including *Colletotrichum gloeosporioides*, Black Sooty Mold, and Leaf Miner (*Acrocercops spp.*).

To improve the generalization capability of the deep learning model and reduce overfitting, data augmentation techniques were applied prior to training. These techniques simulate real-world variations in image capture conditions such as lighting, orientation, and scale.

The augmentation techniques applied include:

- **Geometric Transformations:** Horizontal and vertical flipping to enable the model to recognize disease patterns regardless of leaf orientation.
- **Rotation:** Small-angle rotations (approximately −15° to +15°) to simulate variations in camera positioning during field image capture.
- **Photometric Adjustments:** Modifications to brightness and contrast to account for different lighting conditions encountered in outdoor environments.

These augmentation techniques increase the diversity of the training dataset and improve the robustness of the segmentation model when applied to real-world cinnamon plantation conditions.

2.1.3 Image Acquisition and Preprocessing

The foundation of any highly accurate computer vision and deep learning framework rests entirely upon the quality, consistency, and structural integrity of the input data tensor. Because the primary data acquisition vector is the smartphone camera of a rural cinnamon farmer, the system must aggressively counteract environmental variances, including optical glare from tropical sunlight, dense canopy shadows, and digital sensor noise [1].

To address these field-level data inconsistencies before they reach the YOLO segmentation network, a robust, automated image preprocessing pipeline was engineered using OpenCV (cv2) and NumPy (np) libraries in Python.

- **Image Standardization and Resizing**

First, raw RGB image arrays are geometrically standardized. The YOLO architecture requires a uniform input tensor shape to execute its convolutional operations efficiently. Therefore, all incoming images are programmatically resized to a fixed dimension of 640x640 pixels using standard OpenCV resizing functions. This step ensures perfect tensor compatibility with the pre-trained weights while preserving the necessary macroscopic structural features of the fungal hyphae.

- **Noise Reduction**

In situations where smartphone sensors function under low light conditions or densely foliated areas, noise is introduced both chromatically and luminously. The introduction of this noise causes difficulties when trying to calculate the gradient values for use within the shallow layers of the CNN. In order to remove the noise, this framework utilizes noise reduction techniques that are readily available through OpenCV. The objective here is to reduce the level of noise from the sensors while maintaining clear edge delineation of the lesions.

Figure 6: Methodological pipeline

2.1.4 Spatial Instance Segmentation (YOLO)

The core spatial component of the system is based on the YOLOv8 (You Only Look Once) segmentation model, specifically the Nano variant provided by the Ultralytics library. This model was selected due to its high detection accuracy and lightweight architecture, making it suitable for real-time applications in mobile environments.

The YOLO architecture abandons the traditional, computationally heavy anchor-box paradigm utilized in older object detectors. Instead, it employs a highly advanced anchor-free detection mechanism that directly predicts the center of the fungal lesion. This mathematical shift drastically reduces the total number of bounding box predictions the network must process, significantly accelerating inference speed and vastly improving the network's ability to accurately localize the highly irregular,

chaotic, and asymmetrical geometries characteristic of *Colletotrichum gloeosporioides* and *Stenella spp.* necrotic boundaries

- **Transfer Learning Implementation:** To bypass the need for massive computational training clusters and to rapidly converge on high accuracy, the system aggressively leverages Transfer Learning. The foundational YOLO Nano weights, completely pre-trained on the massive COCO (Common Objects in Context) dataset, were imported into the framework. These pre-trained convolutional filters already possessed a profound mathematical understanding of fundamental visual concepts. During the fine-tuning training phase, these foundational weights were updated using a custom dataset of cinnamon leaf images sourced from Roboflow. The framework connects to an authorized workspace and retrieves the dataset in YOLO format for training. By building upon the pre-trained COCO weights rather than initializing from random values, the network is able to effectively learn the specific visual characteristics of cinnamon leaf diseases, resulting in faster convergence and improved generalization while reducing the risk of overfitting.

2.1.5 Severity Calculation and the IP% Formula

Once the image is pre-processed, it is passed through the fine-tuned YOLO-seg model. The primary output of this model is a highly precise, pixel-level binary segmentation mask. In this mask array, pixels representing the fungal pathogen are assigned a value of 1 (True), while pixels representing healthy tissue and background are assigned a value of 0 (False). Concurrently, a secondary calculation maps the total surface area of the entire cinnamon leaf.

To bridge the gap between computer vision (spatial processing) and time-series forecasting (temporal processing), the raw geometric pixel data must be translated into a structured, continuous numerical feature. This is achieved through the calculation of the **Infection Percentage (IP%)**.

The IP% serves as the primary input feature vector that is sequentially fed into the LSTM/GRU recurrent networks. It provides a mathematically rigorous, normalized representation of disease severity that is entirely invariant to the camera's original focal distance or the absolute physical size of the leaf.

The Infection Percentage is calculated using the following deterministic formula:

$$IP\% = \frac{\left(\sum_{i=1}^{H}\sum_{j=1}^{W} M_{fungus(i,j)}\right)}{\left(\sum_{i=1}^{H}\sum_{j=1}^{W} M_{leaf(i,j)}\right)} \times 100$$

Where:

- $\boldsymbol{IP}\%$ represents the total calculated Infection Percentage.
- **H** and **W** denote the height and width of the image tensor (640x640).
- $\boldsymbol{M_{fungus(i,j)}}$ is the binary segmentation mask generated by YOLO indicating the presence of fungal tissue at pixel coordinate **(i,j)** (Value is 1 if fungus is present, 0 otherwise).
- $\boldsymbol{M_{leaf(i,j)}}$ is the binary segmentation mask indicating the entire surface area of the cinnamon leaf at pixel coordinate **(i,j)**.

By continuously calculating this $\boldsymbol{IP}\%$ over successive days for the exact same leaf, the system automatically generates the crucial multivariate time-series datasets required to effectively train the recurrent neural networks on the pathogen's biological growth velocity.

2.1.6 Recurrent Neural Networks for Spatiotemporal Forecasting

Once the YOLO architecture has successfully extracted the geometric boundaries of the fungal lesion and the IP% severity metric has been mathematically calculated, this spatial data must be analyzed across a chronological timeline to predict the future biological trajectory of the pathogen. Biological diseases are not static; they are highly dynamic, spatiotemporal phenomena characterized by aggressive, non-linear expansion rates driven by complex environmental variables.

To model this complex sequential data, the framework moves from convolutional operations to recurrent operations. The system employs Long Short-Term Memory

(LSTM) networks as the primary, foundational forecasting engine, while simultaneously constructing and training Gated Recurrent Unit (GRU) architectures strictly to serve as a comparative mathematical baseline.

- **Primary Forecasting Engine: LSTM Architecture:**

  Standard Recurrent Neural Networks (RNNs) face limitations due to the vanishing gradient problem when processing long chronological sequences. As the backpropagation algorithm computes gradients across multiple time steps, the error signals can progressively shrink toward zero. As a result, the network struggles to retain information from earlier time steps, reducing its ability to learn long-term dependencies[5].

  To address this limitation, the Long Short-Term Memory (LSTM) architecture is used as the primary forecasting model in the Cinnamon Leaf Fungal Spread Predictor. The key strength of LSTM lies in its introduction of a central memory component known as the Cell State (C_t). This cell state acts as a controlled information pathway that allows important data to flow through the sequence with minimal modification, helping preserve relevant temporal information.
  The LSTM utilizes three main gates to regulate and control the flow of information within the cell state:

Forget Gate ($\boldsymbol{f_t}$):
Determines which information from the previous hidden state (h_{t-1}) and the current input (x_t) should be removed from the cell state. This is achieved using a sigmoid activation function:

$$f_t = \sigma\left(W_f\left[h_{\{t-1\}}, x_t\right] + b_f\right)$$

Input Gate ($\boldsymbol{i_t}$):
Controls which new information should be added to the cell state. A sigmoid layer decides which values to update, while a **tanh** function generates candidate values ($\boldsymbol{\tilde{C}_t}$):

$$i_t = \sigma(Wi \cdot [ht - 1, xt] + bi)$$

$$\tilde{C}_t = tanh\left(W_c\left[h_{\{t-1\}}, x_t\right] + b_c\right)$$

Output Gate (O_t):

Determines the final output hidden state (h_t) by filtering the updated cell state through a $\boldsymbol{tanh}$ function, scaled by a sigmoid gate:

$$o_t = \sigma\left(W_o\left[h_{\{t-1\}}, x_t\right] + b_o\right)$$

$$h_t = o_t \times \tanh(C_t)$$

This gated structure enables the LSTM to capture temporal patterns in fungal disease progression by learning from sequences of Infection Percentage (IP%) values. It allows the model to retain important historical information while adapting to new inputs, making it suitable for predicting future disease spread trends.

- **Comparative Baseline: GRU Architecture:**

In this study, the Gated Recurrent Unit (GRU) is implemented as a comparative baseline to evaluate forecasting performance. GRU is a simplified and computationally efficient variant of the LSTM architecture. It combines the functionalities of the forget and input gates into a single update gate (Z_t) and introduces a reset gate (r_t), while eliminating the separate cell state and relying solely on the hidden state (h_t) for information flow.

Update Gate (Z_t):

Controls how much of the previous hidden state should be retained and passed to the future time step:

Figure 7:Internal gating architecture of the LSTM cell

$$z_t = \sigma\left(W_z\left[h_{\{t-1\}}, x_t\right] + b_z\right)$$

Reset Gate (r_t):

Determines how much of the previous hidden state should be ignored when computing the new candidate state:

$$r_t = \sigma\left(W_r\left[h_{\{t-1\}}, x_t\right] + b_r\right)$$

Candidate Hidden State ($\tilde{h}_t$):

Represents the new information generated based on the reset gate and current input:

$$\tilde{h}_t = tanh\left(W\left[r_t \times h_{\{t-1\}}, x_t\right] + b\right)$$

Final Hidden State (h_t):

Combines the previous hidden state and the candidate hidden state using the update gate:

$$h_t = (1 - z_t) \times h_{\{t-1\}} + z_t \times \tilde{h}_t$$

Due to its simpler structure and fewer parameters, GRU requires less computational power and trains faster compared to LSTM. This makes it suitable for efficient time-series modeling, especially in resource-constrained environments. However, in this study, LSTM is selected as the primary forecasting model due to its stronger capability in capturing longer temporal dependencies in disease progression.

2.1.7 Primary LSTM Architecture Configuration

The engineering of the primary LSTM forecasting engine required meticulous tuning of hyperparameters and layer structures to achieve optimal convergence and prevent the model from aggressively overfitting to the highly specialized temporal dataset. The architecture is constructed using the Keras Sequential API from tensorflow.keras.models.

The network ingests a specifically formatted 3D tensor consisting of [batch_size, sequence_length, features]. For this application, the sequence_length is defined as 5

(representing 5 consecutive historical time-steps of recorded disease severity), and the features dimension represents the calculated Infection Percentage (IP%).

The comprehensive configuration of the primary LSTM network is detailed in Table 2.1.

Table 2:Comprehensive LSTM Architecture Configuration

Layer Index	Layer Type	Units / Neurons	Activation Function	Output Shape	Regularization / Parameters
Input	Sequential Input	N/A	N/A	(None, 5, 1)	Ingests 5-day sequential IP% data.
Layer 1	**LSTM**	128	tanh (internal)	(None, 5, 128)	return_sequences=True (Passes full temporal sequence to next layer).
Layer 2	**Dropout**	N/A	N/A	(None, 5, 128)	rate=0.20 (Stochastically zeroes 20% of neurons to aggressively prevent overfitting).
Layer 3	**LSTM**	64	tanh (internal)	(None, 64)	return_sequences=False (Collapses temporal sequence into a dense feature vector).
Layer 4	**Dropout**	N/A	N/A	(None, 64)	rate=0.20 (Secondary aggressive regularization pass).
Layer 5	**Dense**	32	ReLU	(None, 32)	Rectified Linear Unit applies thresholding to boost non-linearity.

Layer 6	**Dense (Output)**	1	Linear	(None, 1)	Outputs continuous regression variable (Forecasted IP% for target horizon).

- **Optimization and Loss Compilation:**

The deep recurrent network is compiled using the **Adam (Adaptive Moment Estimation)** optimization algorithm. The primary objective function utilized to govern the mathematical backpropagation and gradient descent is **Mean Squared Error (MSE)**. MSE was strategically chosen because it heavily penalizes massive predictive deviations by squaring the error values. Evaluation is tracked using the R^2 Score from sklearn.metrics.

2.1.8 System Architecture and Mobile Integration

The successful, practical deployment of the Cinnamon Leaf Fungal Spread Predictor completely relies upon a highly robust, scalable, and decentralized infrastructure. This system architecture is specifically designed to permanently bridge the massive computational gap between rural Sri Lankan agricultural environments and advanced cloud-based deep learning inference. The architecture is heavily stratified into distinct, highly interoperable operational layers.

1. **The Edge Client Layer (React Native Mobile Application):**
   Serving as the primary, localized human-computer interface, the mobile application is engineered using the React Native framework to ensure native-level performance parity across both Android and iOS operating systems from a single, unified codebase. The edge client is highly optimized for maximum resource efficiency on lower-tier hardware. It facilitates high-resolution image

Figure 8:High-level System Architecture Diagram

capture via the native camera module. The mobile interface provides the crucial user-facing outputs: visualizing the current infection percentage and detailing the actionable recommendations (e.g., pruning, isolating, or spraying) based on the LSTM forecast.

2. **The Cloud Middleware and API Routing Layer:**
   Operating effectively as the central nervous system of the framework, this layer handles all asynchronous HTTP/RESTful API requests originating from the mobile clients. It securely orchestrates the transmission of image payloads from the edge client directly to the heavily fortified inference engines. This layer acts as the absolute gatekeeper, managing security and authentication protocols and interfacing seamlessly with the underlying high-speed database architecture to instantly retrieve the required historical chronological records necessary to feed the recurrent models.

3. **The Heavy Inference Engine and Deep Data Storage Layer:**
   This highly secured, cloud-based stratum houses the massive computational workload of the system. The extremely complex, fine-tuned YOLO instance segmentation models and the primary LSTM recurrent forecasting models reside here. These environments execute rapid, complex tensor inference on incoming data streams to instantly calculate the forecasted fungal spread, returning the deterministic predictions back through the API layer to the user's interface in near real-time.

2.2 Commercialization Aspect of the Product

To ensure that the Cinnamon Leaf Fungal Spread Predictor transitions from an academic prototype into a practical and sustainable agricultural solution, a well-defined commercialization strategy is essential. The successful adoption of this system

depends on its ability to integrate seamlessly into the daily practices of farmers while remaining economically accessible. Therefore, understanding the market environment, identifying the appropriate target users, and defining a suitable deployment model are critical factors for long-term sustainability.

2.2.1 Target Audience

The primary target audience of this system consists of small-scale and medium-scale cinnamon farmers in Sri Lanka, particularly in regions such as Matara, Galle, and Hambantota. These farmers traditionally rely on manual inspection methods and experience-based decision-making. However, increasing challenges such as unpredictable disease outbreaks and climate variability have created a growing demand for intelligent, easy-to-use digital solutions that can support timely and accurate decision-making without requiring advanced technical knowledge.

Secondary target groups include:

- **Large-Scale Plantation Managers:**
  These users require large-scale monitoring and forecasting capabilities across extensive cultivation areas. The system can assist in optimizing resource allocation, scheduling preventive treatments, and reducing overall operational costs.

- **Agricultural Authorities and Extension Services:**
  Government organizations such as the Department of Export Agriculture and the National Cinnamon Research and Training Center (NCRTC) can utilize aggregated prediction data to monitor disease patterns at a regional level. This supports data-driven policymaking, early warning systems, and improved agricultural advisory services.

2.2.2 Market Space

Sri Lanka dominates the global Ceylon cinnamon market, producing over 80% of the world's supply. The export revenue generated by cinnamon cultivation is substantial, presenting a massive and highly centralized addressable market for the SMARTCINNAMON framework. As international quality standards (such as HACCP and Geographical Indication certifications) become increasingly strict regarding pesticide residues, the market demand for precision agriculture tools that can minimize chemical usage through early, targeted intervention is growing exponentially. The predictor occupies a unique, previously empty niche within this market: moving beyond basic diagnostic apps (which are heavily saturated) into actionable, time-series forecasting.

2.2.3 Revenue Earning Strategy

To ensure a balance between rapid user adoption and long-term financial sustainability, the proposed commercialization strategy adopts a tiered Software-as-a-Service (SaaS) model. This approach enables gradual onboarding of users while providing scalable access to advanced system features based on user needs and capacity.

- **Tier 1: Basic Access**
    - **Target:** Smallholder farmers.
    - **Features:**This tier provides access to basic disease detection using YOLO-based segmentation and the calculation of current infection severity (IP%). Users receive general treatment guidance based on detected disease conditions.
    - **Revenue Mechanism:** This tier is offered free of charge to encourage widespread adoption among farmers. It also functions as a data acquisition mechanism, where anonymized user-submitted images can be utilized to improve and retrain the deep learning models over time, enhancing system accuracy and robustness.

- **Tier 2: Pro Access (Subscription Model)**
    - **Target:** Medium to Large Estates and dedicated farming cooperatives.
    - **Features:** This tier provides access to the temporal forecasting component powered by the LSTM model. Users can monitor historical infection trends and receive predictions for future disease progression over multiple time horizons (e.g., 1-day, 3-day, and 7-day forecasts). Additionally, the system delivers dynamic recommendations based on predicted severity levels.
    - **Revenue Mechanism:** Access to this tier is provided through an affordable subscription model, designed to align with local agricultural economic conditions. Subscription packages may be offered on a monthly or seasonal basis, with options for group access through farming cooperatives.

- **Tier 3: Enterprise Analytics (B2B)**
    - **Target:** Government agricultural agencies, research institutions, and large-scale agricultural exporters.
    - **Features:** This tier provides access to aggregated analytical insights through a web-based dashboard. It includes regional disease monitoring, trend analysis, and predictive insights that support large-scale decision-making and policy planning.
    - **Revenue Mechanism:** Revenue is generated through institutional licensing agreements or annual subscription contracts, providing access to advanced analytics and system-level insights.

2.3 Testing and Implementation

This section presents the practical implementation of the Cinnamon Leaf Fungal Spread Predictor. The system is developed using Python by integrating computer vision and deep learning techniques. The implementation focuses on three main components: YOLO-based segmentation, infection severity calculation, and LSTM-based prediction

2.3.1 Code Implementations for the research part

The following code snippets illustrate the essential implementation steps of the system. Only the core logic relevant to disease detection, severity calculation, and prediction is presented.

Environment Initialization and Library Imports

This initial block ensures that all required libraries for computer vision, deep learning, and data processing are properly installed and imported into the Python environment.

Figure 9: Environment Initialization and Library Imports

YOLOv8 Segmentation Model Initialization

This step loads the trained YOLOv8 segmentation model used to detect infected regions in cinnamon leaves.

Figure 10: YOLOv8 Segmentation Model Initialization

Image Processing and Segmentation

This function reads the input image, resizes it to match the model input size, and performs segmentation to extract infected regions.

Figure 11: Image Processing and Segmentation

Infection Percentage (IP%) Calculation

This snippet calculates the infection severity using pixel-based analysis of the segmentation mask.

Figure 12: Infection Percentage (IP%) Calculation

LSTM Model Architecture for Prediction

This snippet defines a simplified LSTM model used to predict future disease spread based on historical infection percentage values.

Figure 13: LSTM Model Architecture for Prediction

Prediction Output Generation

This step generates future infection predictions using the trained LSTM model.

Figure 14: Prediction Output Generation

2.3.2 Testing

To evaluate the performance of the proposed system, the dataset was divided into training, validation, and testing subsets. The model was trained using the training dataset, while the validation dataset was used to monitor performance and prevent overfitting.

The final evaluation was conducted using a separate test dataset to ensure unbiased performance measurement. The testing process simulates real-world conditions where images captured via the mobile application are processed by the system.

The performance of the prediction model was evaluated using standard regression metrics such as Mean Absolute Error (MAE) and prediction accuracy trends. The results confirm that the system can effectively detect fungal infections and provide reliable short-term predictions of disease spread.

2.4 Tools and Technologies

The development and implementation of the Cinnamon Leaf Fungal Spread Predictor were carried out using a combination of modern tools and technologies for image processing, deep learning, and system development.

Programming Language:
Python was used as the primary programming language due to its extensive support for machine learning and computer vision applications.

Deep Learning Frameworks:
TensorFlow and Keras were used to design and train the LSTM model for time-series prediction. These frameworks provide efficient tools for building and optimizing neural network architectures.

Computer Vision Library:
OpenCV (cv2) was used for image preprocessing tasks such as resizing and noise reduction.

Object Detection and Segmentation:
The Ultralytics YOLOv8 framework was used to perform instance segmentation and detect infected regions in cinnamon leaf images.

Dataset Management:
Roboflow was used for dataset preparation, annotation, and augmentation, enabling efficient handling of labeled image data.

Development Environment:
Google Colab was used as the development platform, providing access to GPU acceleration for training deep learning models.

3. RESULTS AND DISCUSSION

The empirical evaluation of the Cinnamon Leaf Fungal Spread Predictor was conducted to validate the effectiveness of transitioning from traditional manual disease monitoring to a data-driven predictive approach. The analysis evaluates both the spatial performance of the YOLO segmentation model, which estimates the current infection severity, and the temporal accuracy of the LSTM and GRU models, which are used to forecast disease progression over time.

3.1 Results

YOLO Segmentation Performance Analysis

The overall performance of the prediction pipeline depends on the accuracy of the YOLO segmentation model in estimating the Infection Percentage (IP%). Errors in segmentation may directly affect the reliability of the forecasting stage.

To evaluate the segmentation performance, the YOLO model was tested across three distinct categories:

- Healthy tissue (non-infected regions)
- Early-stage infected regions (subtle lesions)
- Severely infected regions (clearly visible damage)

The performance was measured using standard evaluation metrics such as Precision, Recall, F1-score, mAP50, and mAP50–95.

Table 3: YOLO Segmentation Performance Metrics

Class	**Precision**	**Recall**	**F1-Score**	**mAP50**	**mAP50-95**
Colletotrichum gloeosporioides	0.926	0.500	0.649	0.533	0.510
Healthy Leaf	0.833	1.000	0.909	0.995	0.895
Average	**0.880**	**0.750**	**0.779**	**0.745**	**0.621**

As shown in Table 3.2, the YOLO segmentation model achieved a Mask mAP50 of 0.745 and mAP50-95 of 0.621 during validation. The model demonstrates good performance in identifying infected regions, while maintaining acceptable accuracy across different disease conditions.

The results further indicate that the model performs differently across classes. The detection accuracy for healthy leaves is significantly higher compared to infected classes, which is expected due to the clearer visual characteristics of non-infected regions. In contrast, early-stage infections present subtle visual differences, making them more challenging to detect accurately.

Additionally, the relatively lower recall value observed for certain infected classes suggests that some infected regions may not be fully captured during segmentation. However, the overall balance between precision and recall confirms that the model maintains reliable detection performance suitable for practical applications.

Figure 15:Segmentation of Black Sooty Mold disease using the YOLO model

The segmentation results show that the model can accurately detect infected regions even when the infection is spread irregularly across the leaf surface.
This behavior is particularly important in real-world scenarios where infection patterns are often irregular and scattered. The ability of the model to detect such patterns ensures that even minor infection areas are identified, enabling early intervention before the disease spreads further.

Figure 16: Segmentation of Colletotrichum gloeosporioides infection

The model also performs well in identifying continuous infected regions and mapping their boundaries accurately.
Furthermore, accurate boundary detection allows for more precise estimation of infection severity. Since the severity calculation is directly dependent on the segmented area, improved boundary accuracy leads to more reliable infection percentage calculations.

Training and Validation Loss Analysis

To evaluate the learning behavior of the predictive models, both training and validation loss values were monitored throughout the training process.

Figure 17: Training and validation loss curves of the LSTM model

The loss curves show a rapid decrease during the initial training epochs, followed by a gradual stabilization. The close alignment between training and validation loss indicates that the model has successfully learned generalizable patterns without significant overfitting.
The smooth convergence of both curves indicates that the selected hyperparameters, including learning rate and batch size, were appropriate for the training process. The absence of large fluctuations in validation loss suggests stable model learning and robustness against overfitting.

Model Performance Visualization (LSTM vs GRU)

To further evaluate predictive performance, the relationship between actual and predicted values was analyzed using scatter plots.

Figure 18: Actual vs predicted infection severity using LSTM model

The LSTM model demonstrates a strong correlation between actual and predicted values, with most data points closely aligned along the ideal regression line. This confirms the model's high predictive accuracy and its effectiveness in modeling disease progression.

Figure 19: Actual vs predicted values using GRU model

In contrast, the GRU model exhibits a weaker correlation, with predictions more widely dispersed from the ideal fit line. This indicates lower predictive accuracy and reduced capability in capturing long-term temporal dependencies.

3.2 Research Findings and Discussion

Spatiotemporal Forecasting Results (LSTM vs GRU)

The primary objective of this study was to move beyond static disease detection and enable accurate prediction of fungal disease progression over time. To achieve this, both LSTM and GRU models were implemented and evaluated.
Based on experimental results, the LSTM model achieved a significantly higher prediction accuracy of 91.1%, while the GRU model achieved approximately 47.2%. This demonstrates that the LSTM model is more effective in capturing complex temporal patterns in infection progression.

Although the GRU model required fewer computational resources and demonstrated faster training, its simplified architecture limited its ability to model long-term dependencies. Therefore, the LSTM model was selected as the primary prediction model in this system.

This difference in performance highlights the importance of model selection in time-series prediction tasks. The ability of LSTM to retain long-term dependencies makes it more suitable for modeling biological processes such as disease spread, which evolve over time. The GRU model, while computationally efficient, may not fully capture such complex temporal relationships.

Multi-Horizon Forecasting Performance

The LSTM model was evaluated for multiple prediction horizons, including 1-day, 3-day, and 7-day forecasts.

The results indicate that prediction accuracy is highest for short-term forecasts and gradually decreases as the prediction horizon increases. Despite this, the model maintains acceptable performance levels even for longer-term predictions.

The evaluation confirms that the model is suitable for short-term disease spread forecasting, which is essential for timely agricultural decision-making.

This trend is expected, as prediction uncertainty increases with longer forecasting horizons. However, the model still provides valuable insights for short-term planning, which is critical for timely intervention in agricultural practices.

Mobile Application Performance

The proposed system was successfully integrated into a mobile-based application to enable real-time disease analysis and prediction.

The average response time from image upload to result generation was approximately **1 second**, demonstrating the efficiency of the system in practical deployment scenarios. This ensures that farmers can receive immediate feedback in real-world conditions.

The fast response time demonstrates the effectiveness of integrating cloud-based inference with mobile applications. This ensures that users can obtain results quickly without experiencing delays, which is essential for real-time decision-making in field conditions.

Discussion and Practical Impact

The results demonstrate that combining image-based disease segmentation with time-series forecasting provides a powerful and effective solution for plant disease management.

The system enables early detection and accurate prediction of fungal disease spread, allowing farmers to take timely preventive measures such as pruning, isolation, and controlled chemical application. This approach not only reduces crop damage but also minimizes unnecessary pesticide usage.

Furthermore, the mobile-based implementation ensures accessibility and usability for farmers in rural environments, making the solution practical and scalable for real-world agricultural applications.

In addition, the proposed system reduces dependency on manual inspection methods, which are often time-consuming and prone to human error. By automating the

detection and prediction process, the system improves efficiency and consistency in disease management practices.

In addition to the observed results, it is important to consider the practical reliability of the prediction system in real-world agricultural environments. Variations in lighting conditions, image quality, and leaf positioning may slightly affect segmentation accuracy. However, the model demonstrates strong robustness under such variations, ensuring consistent performance.

Furthermore, the integration of both spatial and temporal analysis provides a more comprehensive understanding of disease behavior. Unlike traditional systems that focus only on detection, this approach enables predictive insights, which significantly improves decision-making capabilities for farmers.

The results also highlight the importance of high-quality annotated datasets in improving model performance. Increasing dataset diversity can further enhance model generalization and accuracy in future implementations.

4. CONCLUSION

This study successfully developed a mobile-based system for detecting and predicting fungal diseases in cinnamon leaves using a combination of computer vision and deep learning techniques. The proposed system integrates YOLOv8 segmentation for accurate disease identification and severity estimation, along with LSTM-based time-series forecasting to predict future disease progression.

The results demonstrate that the YOLO model can effectively identify infected regions and estimate infection severity under real-world conditions. The LSTM model achieved a high prediction accuracy of **91.09%**, outperforming the GRU model, which achieved **47.17%**, indicating that LSTM is more suitable for capturing long-term temporal dependencies in disease progression.

The integration of the system into a mobile application enables real-time analysis with a response time of approximately one second, making it practical for field-level usage. This allows farmers to make timely and informed decisions regarding disease management, reducing crop loss and minimizing excessive chemical usage.

Overall, the proposed system provides a scalable, efficient, and user-friendly solution for smart agriculture, particularly in cinnamon cultivation. Future improvements may include expanding the dataset, incorporating environmental factors such as humidity and temperature, and enhancing prediction accuracy for long-term forecasting.

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6. APPENDICES

Appendix A: Sample Prediction Outputs

Figure 20: Prediction result for healthy cinnamon leaf

Figure 21: System response for invalid input (non-cinnamon leaf)

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Figure 22:System rejection of unrelated object input (non-leaf image)

Figure 23: Prediction result for infected cinnamon leaf with severity and forecast

Appendix B Turnitin Similarity Report

Figure 24:Turnitin Similarity Report